

Exercise 1:

External API:

```
1 package com.example;
2
3 public interface ExternalApi {
4
5     String getData();
6
7 }
```

My Service:

```
1 package com.example;
2
3 public class MyService {
4     private ExternalApi api;
5
6     public MyService(ExternalApi api) {
7         this.api = api;
8     }
9
10    public String fetchData(){
11        return api.getData();
12    }
13 }
```

MyServiceTest Code:

```
1 package com.example;
2
3 import static org.junit.jupiter.api.Assertions.assertEquals;
4 import static org.mockito.Mockito.*;
5 import org.junit.jupiter.api.Test;
6 import org.mockito.Mockito;
7
8 public class MyServiceTest {
9     @Test
10    public void testExternalApi() {
11        ExternalApi mockApi = Mockito.mock(ExternalApi.class);
12        when(mockApi.getData()).thenReturn("Mock Data");
13        MyService service = new MyService(mockApi);
14        String result = service.fetchData();
15        assertEquals("Mock Data", result);
16    }
17
18 }
```

Output:

```
[INFO] -----  
[INFO] T E S T S  
[INFO] -----  
[INFO] Running com.example.MyServiceTest  
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 1.848 s -- in com.example.MyServiceTest  
[INFO]  
[INFO] Results:  
[INFO]  
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0  
[INFO]  
[INFO] -----  
[INFO] BUILD SUCCESS  
[INFO] -----  
[INFO] Total time: 21.599 s  
[INFO] Finished at: 2026-07-02T17:13:54+05:30  
[INFO] -----
```